Protein Design for SynBio Challenges 2026

by GeneMeow team, Russia

Summary

This final package uses six sfGFP-derived sequences from the DMS-positive/supercharge design axis. The final candidates were selected because they passed competition format checks, showed positive sfGFP-aware brightness delta scores, preserved the chromophore-forming TYG motif, and passed ColabFold core structural validation. The workflow does not claim experimental proof before official testing; it provides a reproducible computational prioritization pipeline.

1. Objective and scoring logic

The objective is to design GFP-family sequences with high initial fluorescence and high post-heat fluorescence retention. In the official assay, each sequence is synthesized, expressed in a cell-free protein synthesis system, measured for initial brightness, heat-treated, and measured again. The combined score is proportional to post-heat brightness relative to the WT control in the same batch:

$$\text{score} = \frac{Finitial}{Finitial\ WT} \times \frac{Ffinal}{Finitial} = \frac{Ffinal}{Finitial\ WT}$$

Therefore, the design strategy must protect initial brightness while increasing the chance of retaining fluorescence after heat treatment.

2. Final computational workflow

- Load official references, GFP brightness data, exclusion list and submission template;
- Use sfGFP as the parent scaffold and preserve the chromophore-forming TYG motif at positions 65-67;
- Evaluate initial brightness using an sfGFP-aware delta model derived from avGFP DMS coefficients, with corrected dataset numbering;
- Evaluate thermal-retention potential using a stability proxy based on negative surface-charge shift and mutation risk;
- Run ColabFold on WT sfGFP and all six candidates to check preservation of the GFP beta-barrel and chromophore environment;
- Submit a portfolio of conservative, balanced and thermal/supercharge candidates.

3. Brightness validation

Brightness was not treated as a directly predicted experimental Finitial value. Instead, each mutation was scored as a delta effect relative to sfGFP using avGFP DMS-derived coefficients. The key correction is that the DMS dataset numbering excludes the initial methionine, so protein position P maps to dataset position P - 1. The numbering check was perfect for the available DMS features.

All six final sequences have positive sfGFP-aware brightness delta scores and all mutation effects used in this validation were measured, not estimated. This is the main advantage of the final DMS-positive design axis.

Seq	Mutations	Brightness delta	Verdict	Measured	Estimated	Ref mismatch	Low support
1	D19E:R73L:H231Y	0.382	>= sfGFP brightness prior	3	0	0	0
2	D19E:R73L:K166E:N212D:H231Y	0.433	>= sfGFP brightness prior	5	0	0	0
3	D19E:R73L:K166E:N198D:N212D:H231Y	0.500	>= sfGFP brightness prior	6	0	0	0
4	D19E:R73L:K101E:K156E:K166E:N198D:N212D:H231Y	0.544	>= sfGFP brightness prior	8	0	0	0
5	D19E:R73L:H231Y:Y237N	0.556	>= sfGFP brightness prior	4	0	0	0
6	K101E:K156E:K166E:N198D:N212D	0.162	>= sfGFP brightness prior	5	0	0	0

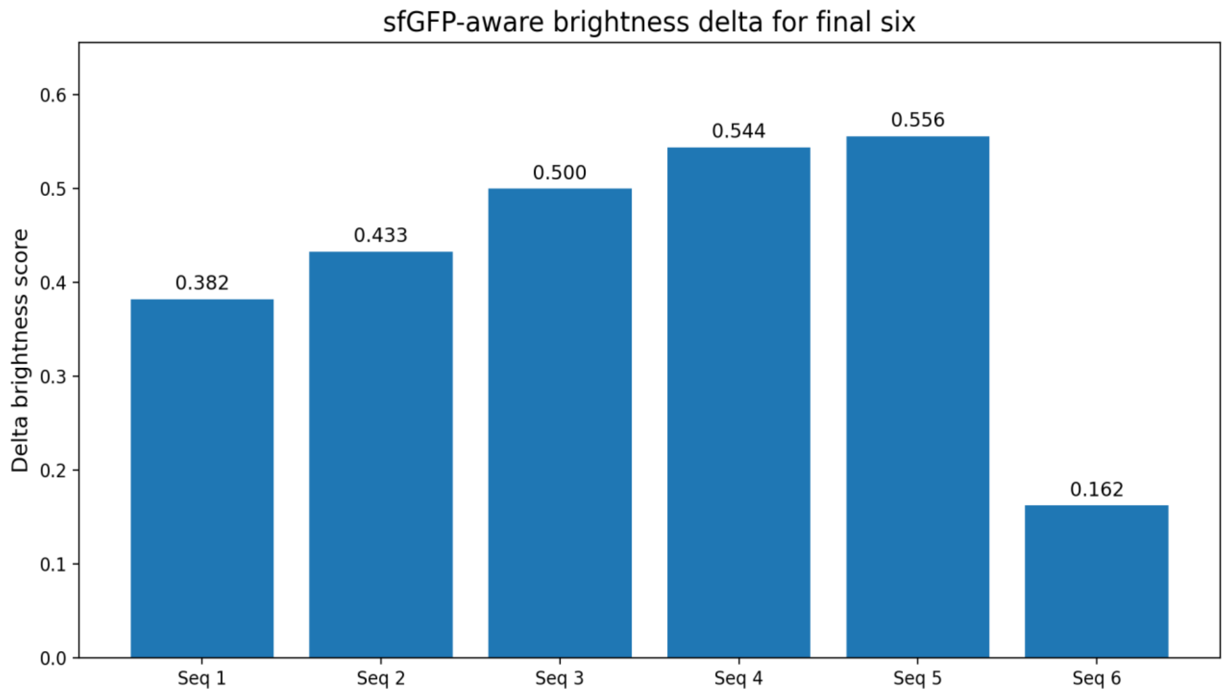


Figure 1. ***All six final sequences have positive sfGFP-aware brightness delta scores.***

4. Stability proxy and risk control

Because no large public dataset exactly matches the official 72C heat-treatment assay, thermal retention was not predicted directly. The workflow used a stability proxy: moderate negative surface-charge shift was treated as a solubility/aggregation-resistance prior, while excessive mutational load was penalized.

The final six include conservative brightness-preserving candidates, balanced charge-shift candidates and stronger supercharge candidates. Seq 3 and Seq 6 are especially strong balanced/thermal candidates by this proxy.

Seq	Charge delta	Stability proxy	Risk penalty
1	1.10	0.316	0.240
2	4.10	1.22	0.400
3	5.10	1.51	0.480
4	9.10	1.30	1.14
5	1.10	0.316	0.420
6	8.00	1.56	0.400

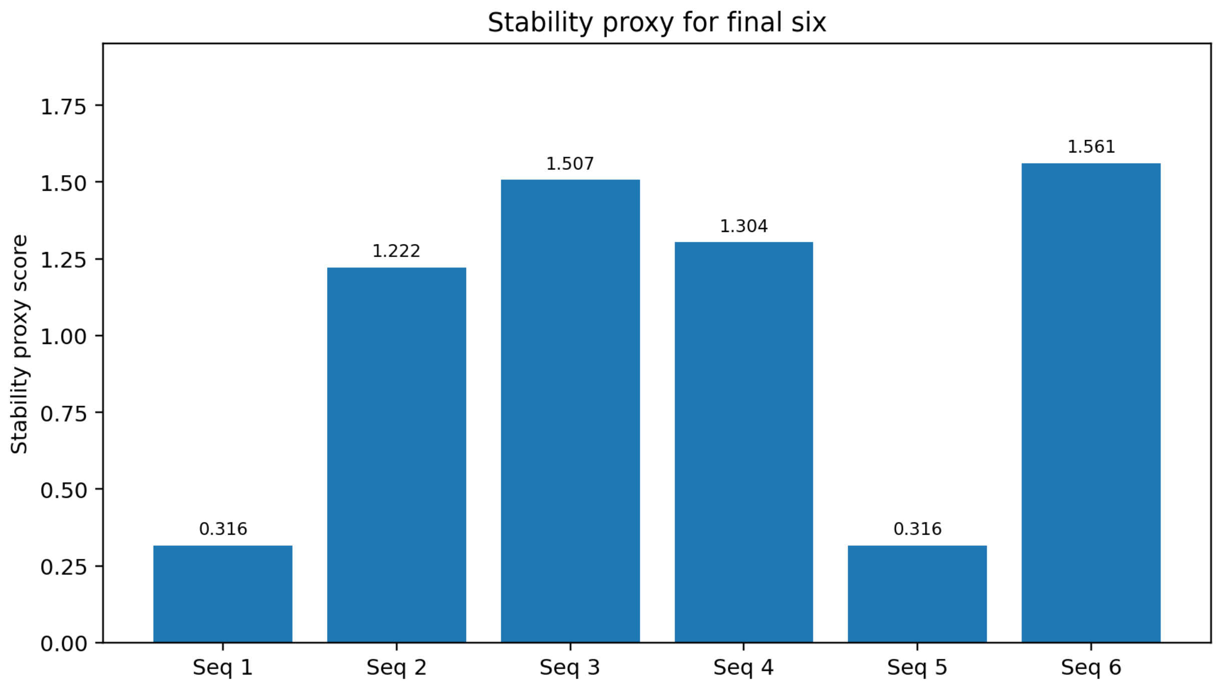


Figure 2. ***Stability proxy values for the final six candidates.***

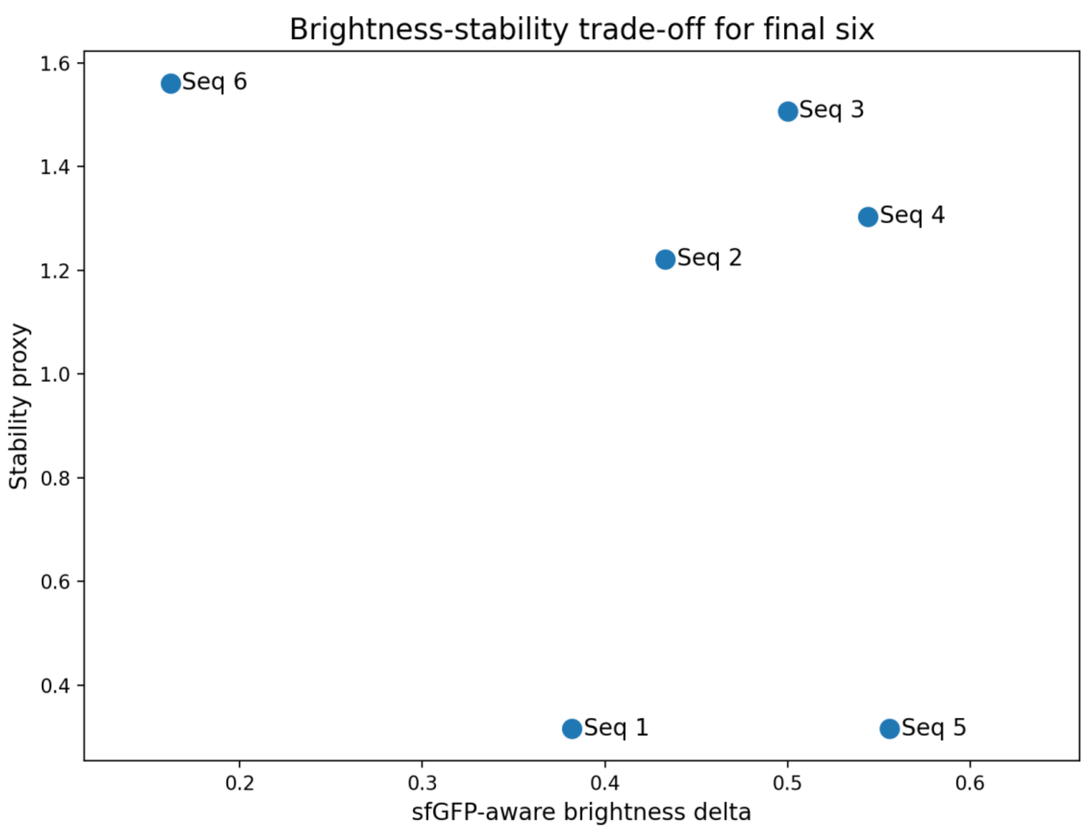


Figure 3. Brightness-stability trade-off across the final six candidates.

5. ColabFold structural validation

WT sfGFP and the six designed variants were evaluated with ColabFold AlphaFold2_batch using MMseqs2 UniRef + Environmental MSAs, five models per sequence, three recycles, no templates and no Amber relaxation. ColabFold was used only as a structural sanity check; it does not predict fluorescence intensity, chromophore maturation or experimental heat retention.

All final candidates preserved a high-confidence GFP-like fold. Mean pLDDT values were about 96, pTM was 0.91 for all variants, chromophore-region confidence was high, and core RMSD to WT sfGFP remained below 0.13 A. Some terminal mutations such as H231Y or Y237N have lower per-position pLDDT because the C-terminus is flexible; the core mutation-site pLDDT remains high in all variants.

Seq	Mean pLDDT	pTM	Chromo min	Core RMSD A	Core mut-site min	Pass
1	96.21	0.910	95.38	0.125	98.12	True
2	96.17	0.910	95.75	0.040	97.75	True
3	96.23	0.910	96.00	0.042	97.12	True
4	96.33	0.910	95.31	0.123	95.88	True
5	96.33	0.910	94.88	0.128	98.12	True
6	96.22	0.910	94.75	0.115	95.75	True

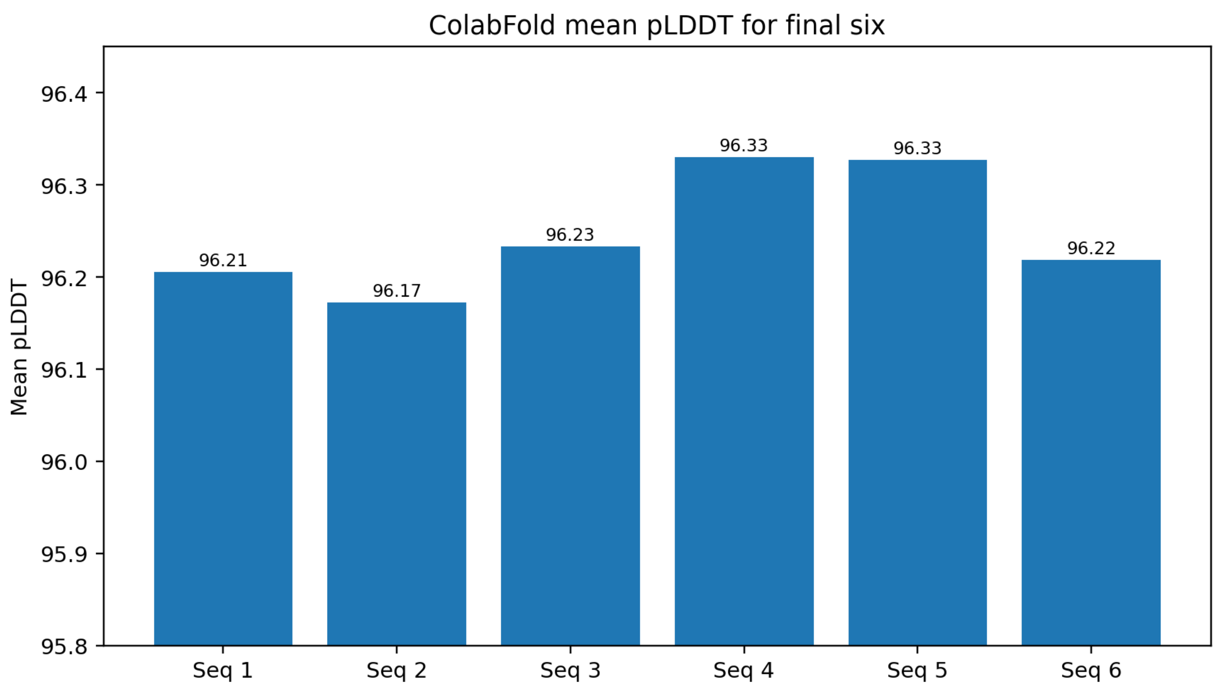


Figure 4. Mean pLDDT for the final six candidates.

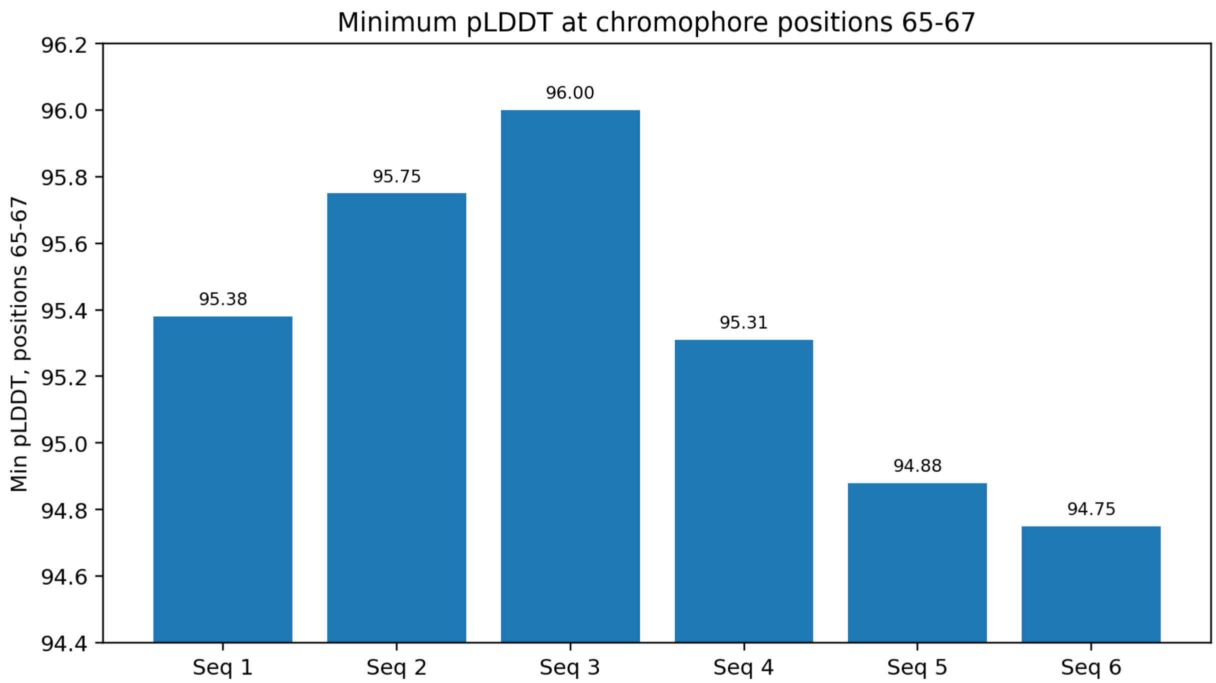


Figure 5. Minimum pLDDT in chromophore-forming positions 65-67

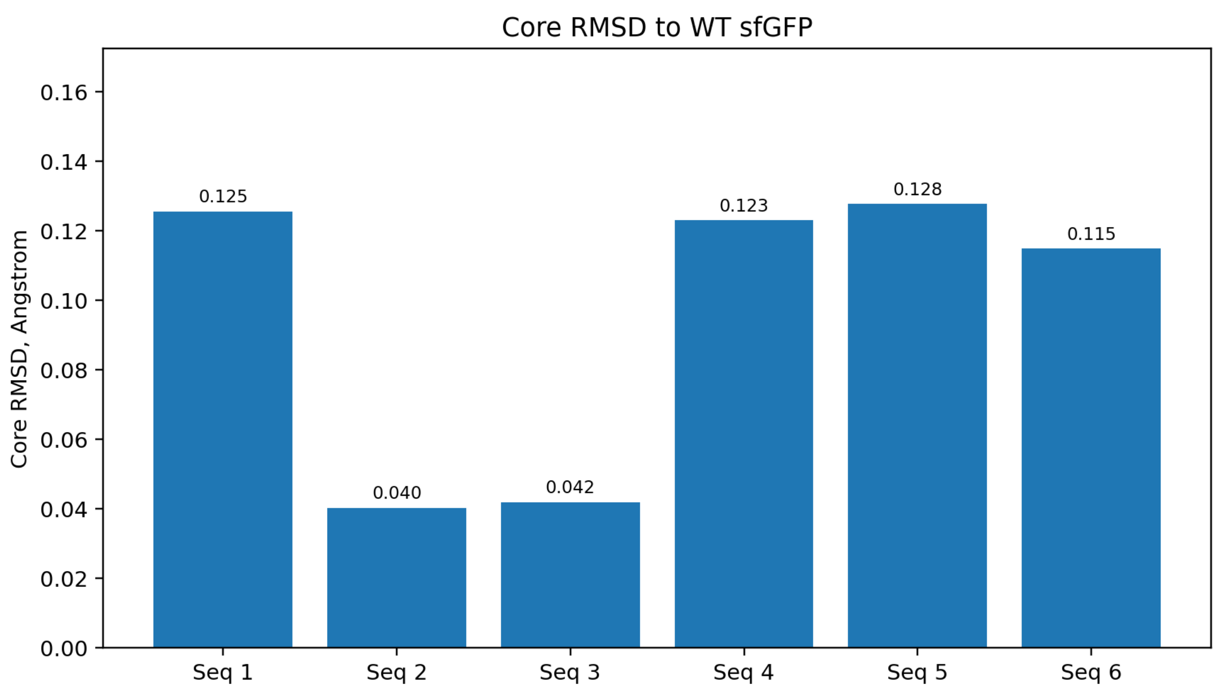


Figure 6. Core RMSD to WT sfGFP.

6. Representative ColabFold output plots

Each design was modeled five times (rank_1–rank_5); the plots below report model *confidence*, not fluorescence or heat behavior.

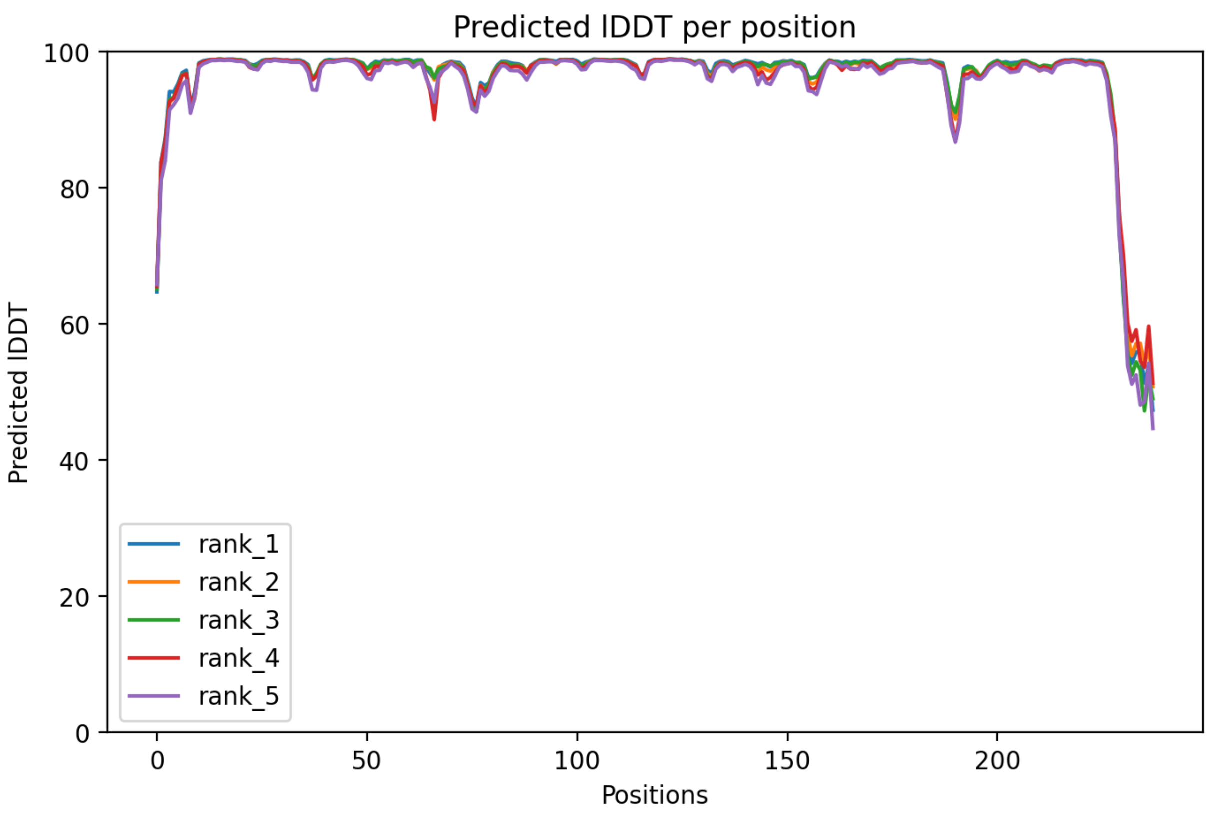
Predicted lDDT per position is per-residue confidence (0–100): it stays ~90–98 across the β-barrel, confirming a confidently predicted fold, while the drop after position ~230 is the intrinsically flexible C-terminal tail (expected for GFP, not affecting the barrel or chromophore).

PAE is the expected error (Å) in the relative position of every residue pair: uniformly low off-diagonal values mean the protein is predicted as one compact, rigid domain, with only the flexible tail showing higher error.

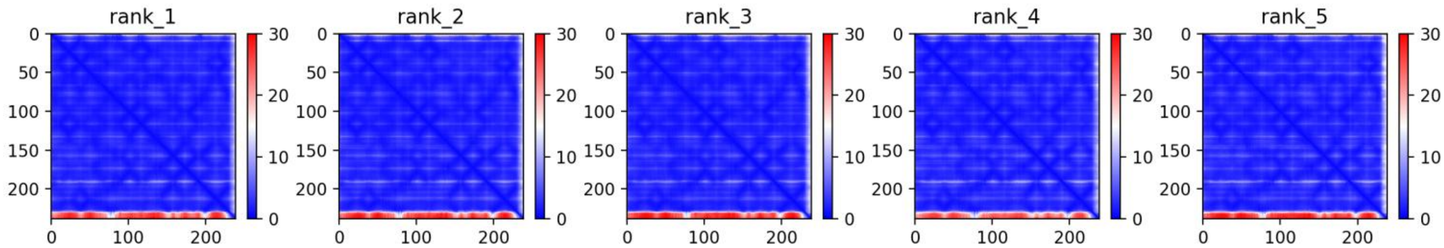
Sequence coverage shows the depth and identity of homologous sequences found per position: the deep coverage (~600+ sequences over most of the length) means the prediction is well-supported by evolutionary information, and the close agreement among the five models is itself a sign of a robust prediction.

None of these predicts brightness, chromophore maturation, or 72 °C retention (see Section 5).

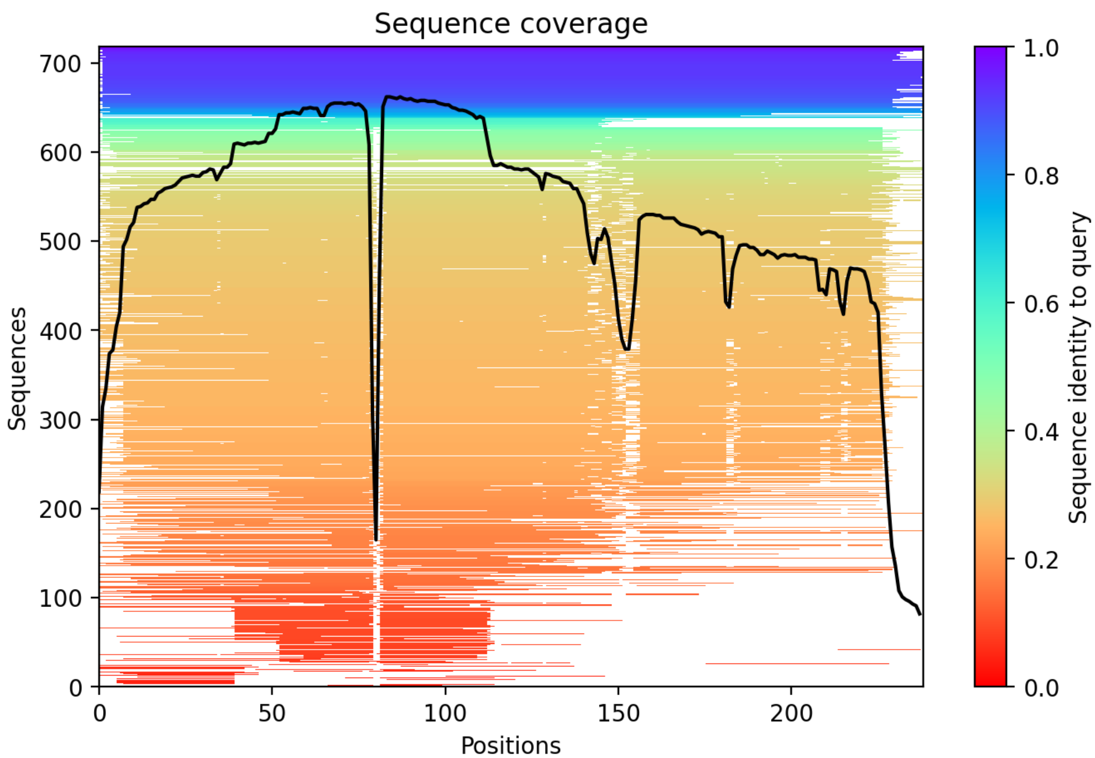
Representative raw ColabFold plots are included for the strongest balanced candidate Seq 3 and the strongest thermal/supercharge candidate Seq 6. The complete raw plots are stored in the GitHub repository.



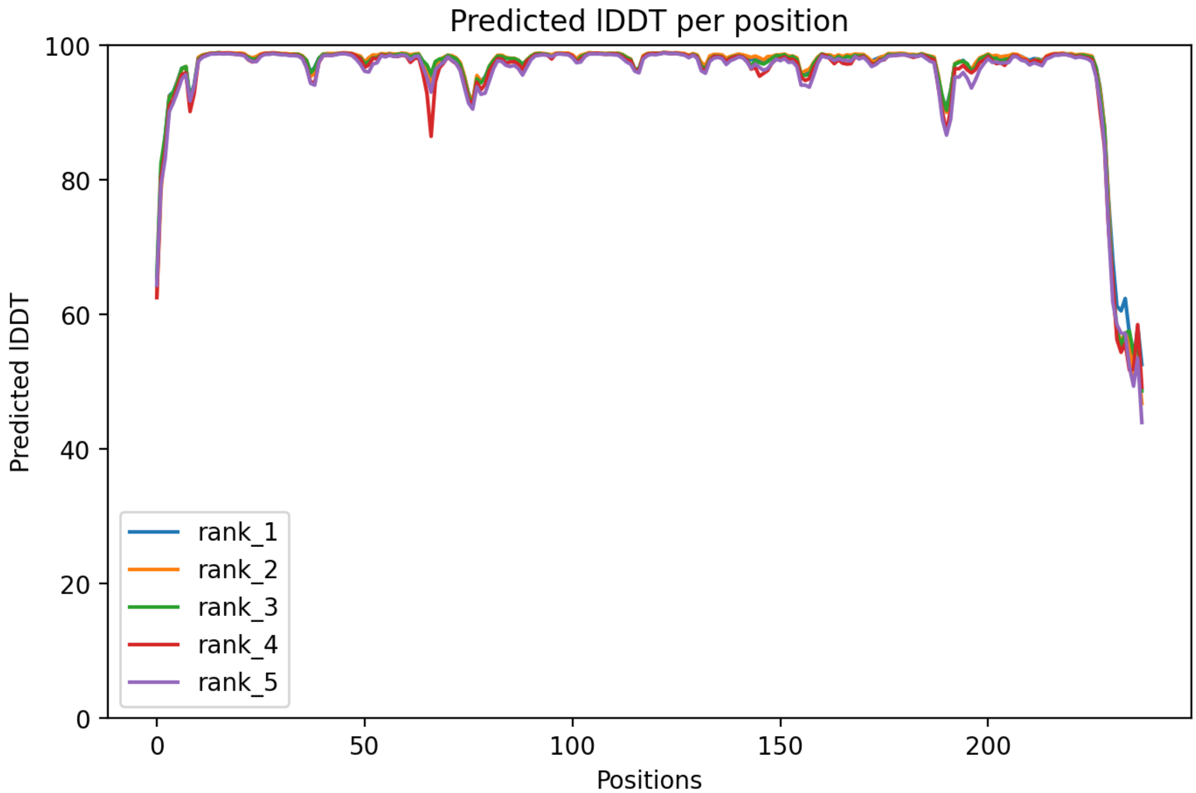
Seq 3 balanced candidate: ColabFold pLDDT plot.



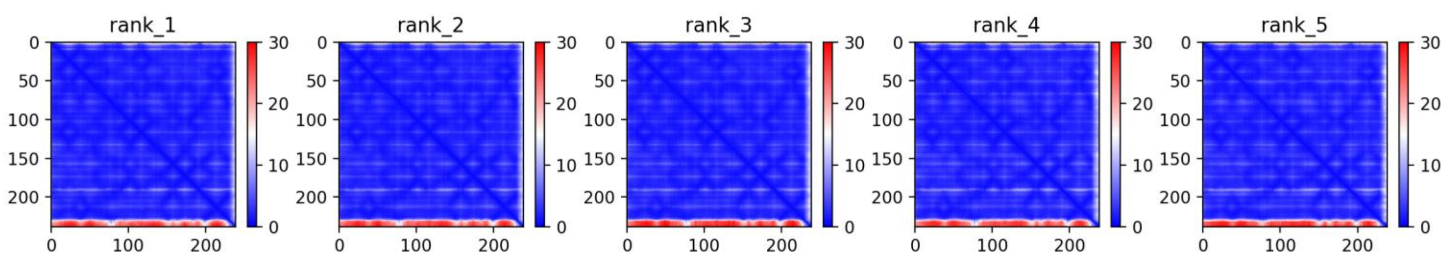
Seq 3 balanced candidate: ColabFold PAE plot.



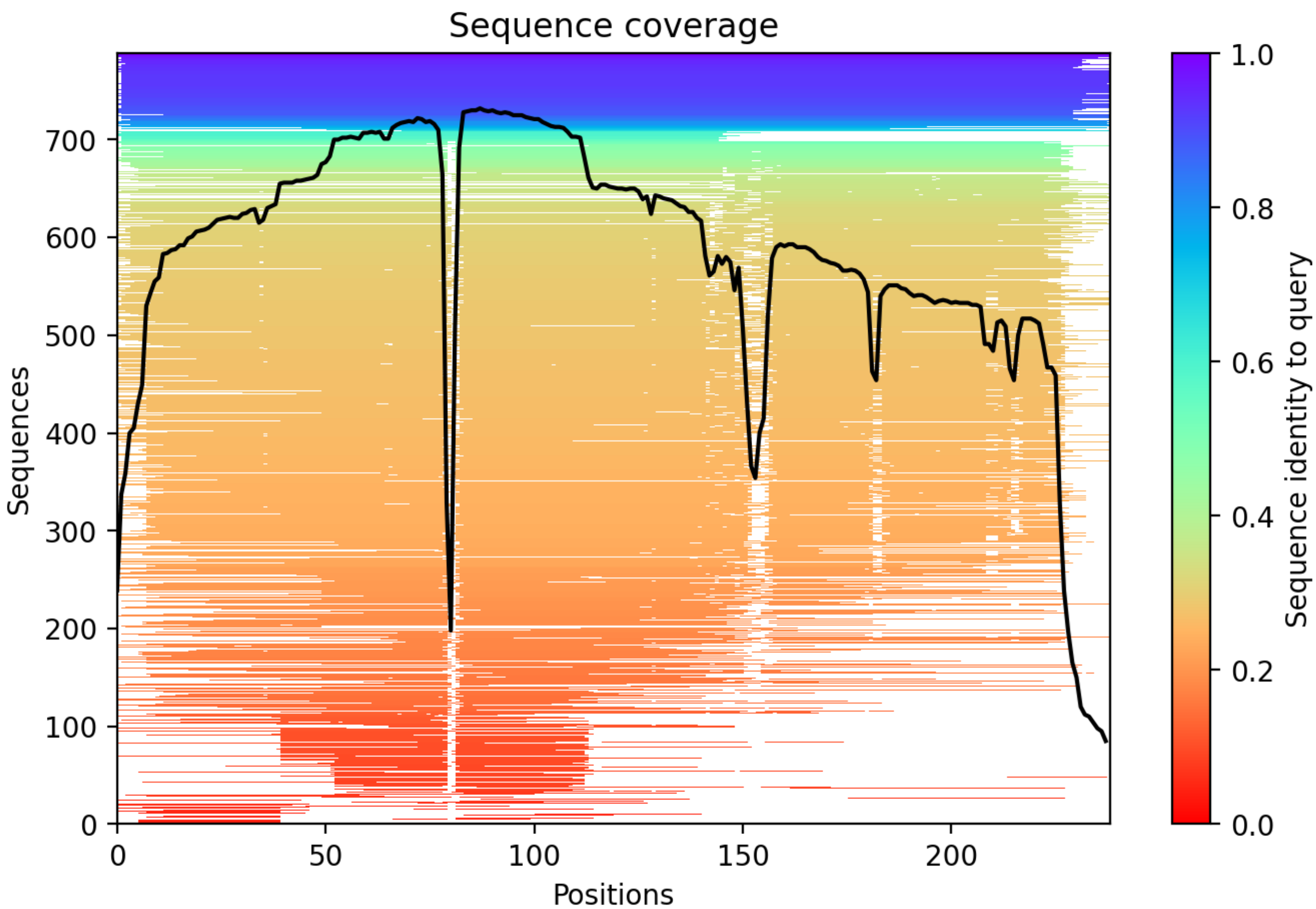
Seq 3 balanced candidate: MSA coverage plot.



Seq 6 supercharge candidate: ColabFold pLDDT plot.



Seq 6 supercharge candidate: ColabFold PAE plot.



Seq 6 supercharge candidate: MSA coverage plot.

7. Final submitted sequences

The submitted CSV contains exactly six rows with the required columns Team_Name, Seq_ID and Sequence. All sequences are 238 aa long, start with M, use only standard amino acids and preserve TYG at positions 65-67.

Seq	Role	Mutations
1	conservative surface brightness	D19E:R73L:H231Y
2	balanced core	D19E:R73L:K166E:N212D:H231Y
3	main balanced	D19E:R73L:K166E:N198D:N212D:H231Y
4	thermal high-risk/high-reward	D19E:R73L:K101E:K156E:K166E:N198D:N212D:H231Y
5	brightness backup	D19E:R73L:H231Y:Y237N
6	supercharge insurance	K101E:K156E:K166E:N198D:N212D

8. Reproducibility package

The accompanying GitHub-ready repository includes the final submission CSV, validation metrics, ColabFold plots, rank-001 PDB/JSON files, scripts for brightness scoring and ColabFold result processing, and this documentation.

9. Limitations

- Brightness delta is a DMS-derived computational prior, not a direct experimental Finitial measurement;
- The stability proxy is not a melting temperature and does not directly predict Ffinal after 72C heat treatment;
- ColabFold validates structural plausibility but does not model chromophore maturation, quantum yield, extinction coefficient, or CFPS expression yield;
- Final performance must be determined by the official experimental assay.

10. LLM Agent usage, logic tree and key execution logs

Large language models and code agents were used as interactive assistants for reasoning, code generation, validation, documentation and artifact packaging. The team retained responsibility for domain decisions, final sequence selection and correction of model outputs. This section provides the compact logic tree and key execution logs required for reproducibility; the complete combined log is included in the GitHub repository as docs/agent_workflow_combined_GeneMeow.md.

10.1 Agent roles and human-in-the-loop controls

- Human team: defined the biological objective, reviewed candidate designs, challenged weak assumptions and approved the final six sequences.
- LLM/code agents worked in two parallel branches. Claude branch: built the sfGFP-aware brightness model (Ridge + isotonic on avGFP DMS), constructed and screened the candidate sequences, and produced the canonical brightness scoring used in Section 3. ChatGPT branch: ran the full ColabFold structural validation (Sections 5-6, Figures 4-6 and the raw plots), built the stability proxy (Section 4), and independently re-built a brightness model used as a convergent cross-check.
- Quality control: every candidate had to pass format checks, exclusion-list screening, sfGFP-aware brightness validation and ColabFold structural sanity checks before being considered for final submission.

10.2 Logic tree

```
GOAL: select six GFP-family sequences for high post-heat brightness
|
+-- Load official competition materials
|   +-- reference GFP sequences, GFP_data.xlsx, Exclusion_List.csv,
submission template
|
+-- Build initial brightness evidence [Claude branch]
|   +-- train Ridge + isotonic model on avGFP brightness DMS data
|   +-- verify DMS numbering offset: dataset_position = protein_position - 1
|   +-- score mutations as delta effects relative to sfGFP, not as absolute
avGFP-to-sfGFP predictions
|
+-- Build stability evidence [ChatGPT branch]
|   +-- use sfGFP as robust scaffold
|   +-- add negative surface-charge / supercharging mutations
|   +-- penalize excessive mutational load and risky designs
|
+-- Generate and audit candidate designs
|   +-- check sequence length, alphabet, initial M and TYG chromophore
preservation
|   +-- remove exact matches to Exclusion_List.csv and previously listed
sequences
|   +-- compare H148S-heavy branch and DMS-positive/supercharge branch
|
+-- Structural validation [ChatGPT branch]
|   +-- run ColabFold AlphaFold2_batch on WT sfGFP and all final candidates
|   +-- extract mean pLDDT, pTM, chromophore pLDDT and core RMSD to WT
|
+-- Final selection and packaging
    +-- choose final six with positive brightness delta and structural pass
    +-- create submission_GeneMeow.csv, design PDF and GitHub repository
```

10.3 Key execution logs (by agent branch)

```
[1] Claude branch - reference and format checks
    sfGFP length = 238 aa; chromophore positions 65-67 = TYG
    Final CSV rows = 6; all sequences start with M; all contain only 20
standard amino acids
    Exclusion_List.csv check = no exact matches

[2] Claude branch - canonical brightness model (Section 3 scoring)
    avGFP brightness rows used = 51715
    mutation features = 1810
    raw Ridge test R2 = 0.6878
    Ridge + isotonic test R2 = 0.933; MAE = 0.160
    numbering offset check = 1810/1810 tokens match avGFP at
protein_position - 1
    final scoring mode = mutation-level delta relative to sfGFP

[3] Claude branch - final six brightness (canonical)
    Seq 1 delta = +0.382
    Seq 2 delta = +0.433
    Seq 3 delta = +0.500
    Seq 4 delta = +0.544
    Seq 5 delta = +0.556
    Seq 6 delta = +0.162
    result = all six have positive sfGFP-aware brightness delta scores

[4] ChatGPT branch - ColabFold structural validation
    number of final sequences evaluated = 6 plus WT sfGFP
    mean pLDDT range = 96.17-96.33
    pTM = 0.91 for all final candidates
    chromophore pLDDT minimum range = 94.75-96.00
    core RMSD to WT sfGFP range = 0.040-0.128 A
    result = all final sequences passed structural sanity checks
```

Note on the two brightness models. The brightness scoring above is the Claude-branch model (1810 features, R2 = 0.933). The ChatGPT branch independently built a second brightness model on the same avGFP DMS data (1703 features, R2 = 0.9214), producing deltas +0.170, +0.283, +0.346, +0.379, +0.279, +0.209 for Seq 1-6. Both models agree that all six designs score >= sfGFP and both verified the -1 numbering offset; they differ in exact magnitude and fine ranking. The higher-validation Claude model is reported as canonical; the ChatGPT model is retained as an independent convergent check. The full ColabFold structural validation in Sections 5-6 was produced by the ChatGPT branch and is used in its entirety.

Data provenance. The brightness model is trained only on the public avGFP DMS dataset (Sarkisyan 2016). The list of past competition winners was used solely for the exclusion check - a design must not duplicate a past winner - never as training data or a design template.

10.4 Self-correction and audit trail

The design process included an explicit self-falsification step. An earlier H148S-heavy GeneMeow branch passed ColabFold but showed weaker brightness-delta evidence. The final DMS-positive/supercharge branch was selected after it showed positive sfGFP-aware brightness delta scores for all six variants while also passing ColabFold structural checks.

Audit step	Observed issue	Correction	Effect on final design
Full-sequence avGFP to sfGFP prediction	Predictions saturated and did not distinguish sfGFP variants	Use mutation-level delta scoring relative to sfGFP	Avoided false absolute brightness claims
H148S-heavy branch audit	Several candidates had negative brightness-delta evidence	Compared against DMS-positive/supercharge branch	Final six changed to the DMS-positive/supercharge set
ColabFold structural audit	C-terminal mutations H231Y/Y237N had lower local pLDDT in flexible tail	Evaluate core mutation-site pLDDT separately from tail pLDDT	No beta-barrel or chromophore disruption detected
Submission package check	Potential risk of bloated archives from full ColabFold result extraction	Use memory-safe packaging and include only essential outputs	Repository remains reproducible and compact

10.5 Reproducibility pointers

- Full agent workflow log: docs/agent_workflow_combined_GeneMeow.md
- Final metrics: outputs/design_report.csv (canonical Claude-branch brightness) and outputs/final6_colabfold_brightness_metrics.csv (ChatGPT-branch brightness cross-check and structural metrics)
- Submission file: outputs/submission_GeneMeow.csv
- ColabFold plots and rank-001 structures: figures/ and models/colabfold_rank001_final6/

Appendix A. Combined agent workflow log (full)

This log documents the agent-assisted workflow for team GeneMeow as a single readable record. It covers both agent branches used by the team: the brightness and sequence-design branch (Claude) and the structural-validation and packaging branch (ChatGPT). It fulfils submission requirement 2(2): display the Agent's logic tree and key execution logs so the process is reproducible.

Agent and roles

• Brightness / sequence branch: Claude (Anthropic), used as an interactive reasoning and code agent.

• Structural / packaging branch: ChatGPT (OpenAI), used as a reasoning, code-debugging, validation and packaging agent.

• Role split (honest): the team (human) owns the domain expertise, the decisions, and the critical correction; the agents do the analysis, the code, and the computation. Human corrections of agent mistakes are part of the record below, because the human-in-the-loop self-correction is the method.

• Reproducibility: every brightness number in this log is regenerated by python gfp_pipeline.py plus the short diagnostics snippet in C2. No figure here is hand-entered.

A. Logic tree (brightness / sequence branch, Claude)

```
GOAL: six GFP-family sequences maximizing post-heat brightness   score =
F_final / F_initial_WT
|
+-- [frame]             Score algebra: (F_init/F_init_WT) x (F_final/F_init) =
F_final/F_init_WT.
|                       GFP brightness is stability-gated (Sarkisyan 2016
threshold), so BOTH
|                       factors are stability read-outs at two temperatures
(initial, 72 C).
|
+-- [decision, human]  Anchor on sfGFP: robust fold gives high F_initial plus
a large stability
|                       margin to spend on heat retention.
|
+-- AXIS 1, initial brightness
|   +-- [agent]             The only large GFP brightness DMS is avGFP
(Sarkisyan, 51,715
|   |                       variants) -> additive Ridge + isotonic. Held-out R2
= 0.933.
|   +-- [agent, footgun]   Numbering: the avGFP DMS drops the initial Met, so
|   |                       dataset_pos = protein_pos - 1. Verified (1810/1810
measured
|   |                       tokens match avGFP at offset -1); the pipeline
ASSERTS this and
|   |                       refuses to run otherwise. A wrong offset silently
flips signs.
|   +-- [agent + human]    Brightness drivers are dual-validated (positive on
BOTH the Ridge
|                           coefficient AND the single-mutant DMS brightness) -
```

```
> high-coefficient
|                           but co-occurrence-inflated mutations were rejected.
|
+-- AXIS 2, heat retention
|   +-- [decision]          Negative surface supercharging (Thermal Green
Protein, Close 2015):
|                           surface K/N -> E/D, brightness-neutral in the DMS,
far from the
|                           chromophore. Exclude the epistatic pair N198D +
K214E (-0.18 in DMS).
|
+-- RELIABILITY SCREEN
|   +-- [agent]             Four independent signals per mutation: DMS support,
additivity
|                           residual, burial (rSASA), chromophore distance. ->
flagged L44M
|                           (buried, ~7.4 A) and replaced it with the clean
surface driver
|                           Y237N (concrete design change).
|
+-- DATA PROVENANCE
|   +-- The model is trained only on the public avGFP DMS dataset (Sarkisyan
2016). The list
|       of past competition winners was used solely for the exclusion check (a
design must not
|       duplicate a past winner), never as training data or a design template.
|
+-- MODEL CALIBRATION AND SCOPE
|   +-- [agent discovery]  avGFP -> sfGFP transfer is epistatic: S65T coef = -
0.349 although
|   |                       sfGFP is brighter; the net background puts sfGFP AT
the calibration
|   |                       ceiling -> the model cannot predict "brighter than
sfGFP", only
|   |                       "not dimmer". Therefore score by DELTA relative to
sfGFP.
|   +-- [human correction] An earlier "chromophore wall" claim was overstated
-> corrected to
|                           "high-variance region": deleterious mutations are
enriched at
|                           chromophore-internal residues, yet the chromophore
is also the
|                           engineer's best lever. Tools blind to the
chromophore get burned
|                           there; surface engineering is safe.
|
+-- CORRECTED, sfGFP-AWARE MODEL  [agent build]
|   +-- Verified and asserted offset; delta-not-absolute scoring; reference-
residue check
|       (a mutation is scored directly only if its "from" residue matches both
sfGFP and
|       avGFP); coverage-aware fallback (only ~40% of single substitutions are
measured;
|       unseen -> position-level estimate plus flag, never a silent zero);
honest confidence
|       flags (measured / estimated / ref-mismatch / low-support /
```

```
buried+near-chromo);
|       brightness-only scope (does NOT predict 72 C retention).
|
+-- MERGE  (team decision across the two branches)
|   +-- [decision]         One full workflow: the corrected brightness gate, 
the clean
|                          surface-validated sequences, and the structural 
branch's ColabFold
|                          figures and memory-safe packaging. The final six 
pass the brightness
|                          gate cleanly (delta >= 0, all measured, none 
chromophore-proximal).
|
+-- CONSOLIDATION  [agent]
    +-- Collapsed the modular code into one self-contained file 
gfp_pipeline.py; verified it
        produces byte-identical submission.csv and design_report.csv.
```

Self-correction log (agent error, human catch, fix)

#	Agent's initial output	Caught by	Correction
1	Absolute avGFP -> sfGFP brightness transfer	self / human during model calibration	S65T epistasis plus sfGFP at calibration ceiling, so switched to delta vs sfGFP; this became the shipped model
2	"Chromophore wall" (mutations near the chromophore always bad)	human ("are there papers? always?")	Corrected to "high-variance, high-reward region"; the chromophore is the engineer's best lever, but tools blind to it get burned

These corrections are why the final model scores by delta, asserts its numbering, flags unseen and ref-mismatch mutations, and why the portfolio is deliberately framed as a high floor, not a claim to beat sfGFP on brightness.

B. How to reproduce

```
pip install -r requirements.txt
# place the organizer files in data/ (see data/README.md), then:
python gfp_pipeline.py          # LOG C1 below: model + screen + gate + export
```

The diagnostics in C2 are produced by importing GFPDesignModel from gfp_pipeline.py. All values match the shipped outputs/design_report.csv.

C1. Key execution log (full pipeline run)

```
[1] Brightness model: held-out R^2 = 0.933 (ridge alone 0.688) | offset 
1810/1810 at -1 | sfGFP at ceiling (3.869=3.869) -> rank by delta vs sfGFP
[2] Exclusion list: 135412 sequences | past winners: 20
[3] Building 6 designs (sfGFP net charge -6):
    Seq 1 [conservative_surface  ] 3 mut | delta vs sfGFP +0.382 (3m/0e/0rm,
```

```
0nc) | net -7
    Seq 2 [balanced_core         ] 5 mut | delta vs sfGFP +0.433 (5m/0e/0rm,
0nc) | net -10
    Seq 3 [MAIN_balanced         ] 6 mut | delta vs sfGFP +0.500 (6m/0e/0rm,
0nc) | net -11
    Seq 4 [BEST_thermal          ] 8 mut | delta vs sfGFP +0.544 (8m/0e/0rm,
0nc) | net -15
    Seq 5 [BEST_brightness       ] 4 mut | delta vs sfGFP +0.556 (4m/0e/0rm,
0nc) | net -7
    Seq 6 [INSURANCE_supercharge ] 5 mut | delta vs sfGFP +0.162 (5m/0e/0rm,
0nc) | net -14
[3b] Reliability screen (global additive MAE 0.147): Seq 1-6 all clean
[3c] Brightness gate (delta >= 0, all measured, no chromophore-proximal): Seq
1-6 all pass
[4] Wrote outputs/submission.csv  (Team_Name = GeneMeow)
[5] Wrote outputs/design_report.csv
Done.
```

C2. Key execution log (model diagnostics, guards, audit, self-test)

```
held-out R^2 .................. 0.933
numbering offset check ........ 1810/1810 tokens match avGFP at -1
DMS coverage ................. 40% of single substitutions
S65T coefficient ............. -0.349  (negative although sfGFP is brighter ->
epistatic)
sfGFP absolute prediction .... 3.869  (= ceiling 3.869 -> cannot predict
'brighter than sfGFP')

sanity / guard behaviour:
  S205Y (chromophore proton-wire): delta -1.016 -> below WT (predicted dimmer
by ~1.02); flags ['buried+near-chromo']
  D190W (unseen): measured=False, fallback flags ['estimated'], delta -0.005
  A30E (wrong 'from' residue): rejected -> parent has R at position 30, not A
  R30E (sfGFP R30 vs avGFP S30): flags ['ref-mismatch', 'estimated']

earlier H148S candidate set (6) scored through the SAME corrected gate:
  #1: delta -0.470 -> RE-PICK (1 estimated, 2 buried+near-chromo)
  #2: delta -0.449 -> RE-PICK (1 estimated, 2 buried+near-chromo)
  #3: delta -0.414 -> RE-PICK (1 estimated, 2 buried+near-chromo)
  #4: delta -0.382 -> RE-PICK (1 estimated, 2 buried+near-chromo)
  #5: delta -0.393 -> RE-PICK (1 estimated, 2 buried+near-chromo)
  #6: delta -0.974 -> RE-PICK (0 estimated, 2 buried+near-chromo)
    L220V: effect -0.166 measured=True ['low-support', 'buried+near-chromo']
    H148S: effect -0.367 measured=False ['estimated', 'buried+near-chromo']
    Q183R: effect -0.924 measured=True ['buried+near-chromo']
```

Interpretation. C2 shows the guards firing as intended: a chromophore proton-wire mutation (S205Y) is read as strongly dimming; an unseen mutation falls back instead of scoring zero; an invalid "from" residue is rejected; and a position where sfGFP differs from avGFP is flagged. The same gate rejects the team's own earlier H148S candidate set (those candidates rest on the chromophore-region anchors L220V / H148S / Q183R), while the final six pass cleanly.

D. Role and scope of the structural / packaging branch (ChatGPT)

• Agent: ChatGPT (OpenAI), used as an interactive reasoning, code-debugging, validation and packaging agent.

• Human-in-the-loop role: the user supplied official files, the other branch's outputs, ColabFold result archives, and correction requests, and made the final decision to adopt the DMS-positive surface/supercharge set after the comparative evidence was reviewed.

• Scope: format validation, brightness and stability scoring interpretation, ColabFold archive parsing, comparison of candidate sets, final sequence selection, and PDF / README / GitHub packaging.

E. Logic tree (structural / packaging branch, ChatGPT)

```
GOAL: produce a clean final submission package: CSV + PDF + GitHub repository
|
+-- [initial route] Build an early candidate set
|   +-- Train avGFP DMS brightness model: Ridge + isotonic, R2 ~ 0.9214
(independent build)
|   +-- Discover full-sequence avGFP -> sfGFP transfer saturation
|   +-- Switch to mutation-level, sfGFP-aware brightness sanity checks
|   +-- Generate an early H148S / supercharge candidate set and package an
early submission
|
+-- [adopt route] Brightness-branch DMS-positive surface/supercharge set
|   +-- Validate format: 6 sequences, 238 aa, M-start, standard amino acids
|   +-- Score brightness with the sfGFP-aware delta model -> all six positive
vs sfGFP
|   +-- Score thermal / stability proxy -> sequences 3 and 6 strongest by
proxy
|   +-- Parse the ColabFold archive -> all six pass core structural validation
|   +-- Compare against the earlier H148S candidate set -> the
surface/supercharge set is
|       stronger on brightness and similar or better on structure
|
+-- [final decision] Use the DMS-positive surface/supercharge set as the final
submission
    +-- Rebuild submission_GeneMeow.csv
    +-- Rebuild the design document with plots and ColabFold images
    +-- Rebuild the GitHub-ready archive with standardized filenames
    +-- Rewrite README to match the final data
```

F. Key execution log (structural / packaging branch, ChatGPT)

```
[1] Loaded official and working artifacts:
    - AAseqs of 5 GFP proteins, GFP_data.xlsx, Exclusion_List.csv,
submission_template.csv
    - the brightness-branch submission.csv
    - the ColabFold result archive
[2] Validated final submission table:
    rows: 6 | Team_Name: GeneMeow | Seq_ID: 1..6 | length: 238 aa
    checks: M-start, standard amino acids, TYG chromophore at 65-67, no
duplicated sequences
[3] Re-scored the six sequences with the sfGFP-aware brightness delta model:
    all six have positive delta vs the sfGFP brightness prior; all six pass
the brightness gate
[4] Recomputed stability proxy and assigned roles:
    Seq 1 and Seq 5 conservative/brightness; Seq 2 and Seq 3 balanced;
    Seq 4 higher-risk/high-reward thermal; Seq 6 insurance supercharge
[5] Parsed ColabFold rank-001 outputs:
    Mean pLDDT 96.17-96.33 | pTM 0.91 for all | chromophore pLDDT min 94.75-
96.00 |
    core RMSD to WT 0.040-0.128 A | all pass structural validation
[6] Terminal mutation interpretation:
    H231Y and Y237N show lower local pLDDT (flexible C-terminus); core
mutation-site
    pLDDT remains high, so this was not treated as barrel failure
[7] Final packaging:
    created submission_GeneMeow.csv, the design document, and the GitHub-ready
archive;
    standardized output filenames and paths; updated README to the final
design set
```

G. Final six (combined brightness, stability and ColabFold validation)

Seq	Mutations	Brightness delta (Claude, canonical)	Charge delta	Stability proxy	Mean pLDDT	pTM	Chromophore min pLDDT	Core RMSD A	Structural pass
1	D19E:R73L:H231Y	0.382	1.1	0.3161	96.21	0.91	95.38	0.125	True
2	D19E:R73L:K166E:N212D:H231Y	0.433	4.1	1.2219	96.17	0.91	95.75	0.040	True
3	D19E:R73L:K166E:N198D:N212D:H231Y	0.500	5.1	1.5072	96.23	0.91	96.00	0.042	True
4	D19E:R73L:K101E:K156E:K166E:N198D:N212D:H231Y	0.544	9.1	1.3036	96.33	0.91	95.31	0.123	True
5	D19E:R73L:H231Y:Y237N	0.556	1.1	0.3161	96.33	0.91	94.88	0.128	True
6	K101E:K156E:K166E:N198D:N212D	0.162	8.0	1.5609	96.22	0.91	94.75	0.115	True

Brightness delta is the canonical Claude-branch model (held-out R2 = 0.933). The ChatGPT-branch independent model gave +0.170, +0.283, +0.346, +0.379, +0.279, +0.209; both models agree that all six designs score at or above sfGFP and both verified the -1 numbering offset, differing only in exact magnitude and fine ranking.

H. Final artifact log

```
Final CSV ................. submission_GeneMeow.csv
Final design document .... design_concept_GeneMeow.docx / .pdf
Final repository archive . gfp_2026_genemeow_design_final.zip
Canonical metrics ........ outputs/design_report.csv
Structural + cross-check . outputs/final6_colabfold_brightness_metrics.csv
```

I. Consolidated interpretation

The two agent branches converged on the same final principle: do not claim experimental improvement before the official assay. Both independent brightness models agree that all six designs score at or above sfGFP and both verified the -1 numbering offset. The model is trained only on the public avGFP DMS dataset (Sarkisyan 2016); the list of past winners was used only for the exclusion check, never as training data. Instead of an improvement claim, the team submits a rationally prioritized six-sequence portfolio with positive sfGFP-aware brightness delta evidence, a negative surface-charge supercharging stability proxy, a preserved chromophore-forming motif, high-confidence ColabFold structural validation, and a clear human-in-the-loop correction history. The final submission is presented not as experimentally validated, but as a reproducible computational prioritization workflow optimized for the official dual-objective score.